Year 10 Science
Energy worksheet

1. A dog weighing 9kg is sitting on top of a cliff 17m high. What is the dog's potential energy?

2. How much potential energy is stored when a weight lifter lifts 150kg over his head (1.9m total)?

3. A feather (mass 2g) and a rock (mass 2kg) are held 150m above the ground, which has greater potential energy and by how much?

4. How high is a box mass 5kg above the ground to have a potential energy of 498J?

5. How much potential energy does an object mass 50g held 160cm above the ground have?

6. A bike rider of mass 80kg is traveling down a road at $0.15ms^{-1}$. Calculate the rider's kinetic energy.

7. Sarah was driving along the road in her car (total mass 1.8 tonne) at a speed of $17.5ms^{-1}$. Calculate the kinetic energy of her car.

8. An athlete of mass 94kg completed the 100m race in 10 seconds. Calculate the kinetic energy of the athlete.